Multilabel Aerial Image Classification With a Concept Attention Graph Neural Network

Dan Lin, Jianzhe Lin, Liang Zhao, Z. Jane Wang, *Fellow, IEEE*, and Zhikui Chen, *Member, IEEE*

***Abstract*— Compared with natural images, aerial images collected by satellite sensors/aerial cameras can provide a much larger field of view and often contain multiple objects of interest (multiple labels). There are certain limitations of existing multilabel aerial image classification methods. First, label correlations were often ignored in previous MAIC work, and thus, multilabel classifiers failed to be self-adapted. Second, existing multilabeled data sets for aerial images only cover limited images with fixed labels. Therefore, the underlying semantic correlations of labels cannot be fully included, while such correlation information is implicitly used as common knowledge by human beings. To tackle these concerns, we propose a novel multilabel classification method for aerial images. Our contributions are twofold. First, as the first attempt, label correlations are inferred from both the specific data set and ConceptNet (a popular knowledge graph for common sense). Second, based on graph neural network (GNN), we propose a novel end-to-end aerial image classification model, named the multiple label concept graph (ML-CG). ML-CG builds a concept graph to describe the semantic correlations from both the label set and the ConceptNet. We also incorporate both semantic attention and label attention in the GNN to better extract meaningful information of image labels. Compared with state-of-the-art methods, the effectiveness of the proposed method is demonstrated on both the commonly used UCM data set and a recently proposed DFC15 data set with high image resolution.**

***Index Terms*— Aerial images, graph neural network, multilabel image classification.**

I. Introduction

AERIAL image classification has attracted interests in various applications, including urban mapping and traffic management [1], ecological environmental monitoring [2], and geomorphological analysis [3], [4]. Current classifiers focus mainly on single-label aerial image classification (SAIC) [5]. SAIC aims to map a given image to a single object (label).

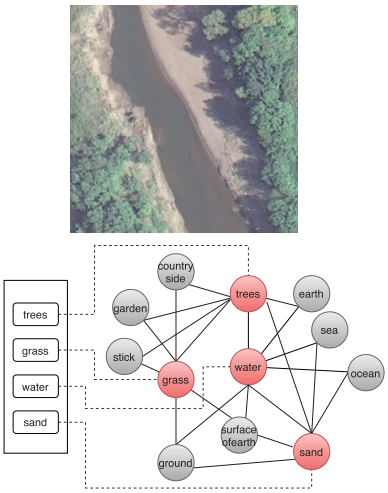


Fig. 1. Example multilabel aerial image with its concept graph in ConceptNet. (a) Image (with the grass, sand, tree, and water objects). (b) Concept graph of the labels of this image.

However, due to the advances in remote sensing, many currently available aerial images are with a much larger field of view and, thus, are inherently multilabel. As shown in Fig. 1, several objects exist in the illustrative river image, including *grass*, *trees*, *sand*, and *water*. To this end, the emerging multilabel aerial image classification (MAIC) task provides a more fine-grained and image-specific understanding for subsequent aerial image learning problems.

Many MAIC methods have been developed to extract representative aerial image features and identify multiple objects. Earlier methods utilized traditional approaches, such as stacked sparse auto-encoders [6] and conditional random field (CRF) [7], to decode images into a low-dimensional feature space. Recent deep learning methods can learn image features in a trainable end-to-end manner and show promising representations [8], [9]. However, there are certain design limitations in the existing deep learning methods, as described follows.

First, most existing methods ignored the correlations between the objects in aerial images. Although explored widely in common computer vision tasks using natural images [9], [10], label correlations are rarely studied in MAIC. Second, in a few existing approaches that explored label dependencies, label correlations were predefined and fixed during label predicting. Due to such assumed static label correlations, multilabel classifiers failed to be self-adapted. Third, in the aforementioned approaches that do consider label

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dependencies, label correlations were only extracted within the specific data set (for the study of interest). These approaches cannot take advantage of additional knowledge of the underlying semantic correlations in real-world images (from other data sources) though such information is common knowledge for human beings. Motivated by these observations, we develop a deep learning approach that leverages comprehensive label correlations from not only the specific data set but also human knowledge from existing knowledge graphs.

Recently, knowledge graphs were proposed to represent concepts and semantic correlations in a graph structure, and they hold strong power for knowledge reasoning and supplement in many neural language processing problems [11]. Existing MAIC models define label correlations by the coexistence of multiple labels in image samples. Differently, we build a concept graph for the whole label set utilizing the human knowledge in ConceptNet [12], a real-world knowledge graph of human concepts. For example, a concept graph for labels existing in Fig. 1 is illustrated in Fig. 1. The red nodes are the objects labeled in the image, and the gray nodes indicate the related objects of red nodes in ConceptNet. Such a graph can introduce more objects to characterize the underlying correlations among labels.

In the concept graph, we define two types of label correlations between two labels from the label set, explicit correlations and implicit correlations. The former represents that two label entities adjoin with each other in a triple in the ConceptNet (seen the node *trees* and node *water* in Fig. 1). The latter represents that two label entities connect indirectly by both adjoining with a third common entity in the knowledge graph (seen the node *grass* and node *water* in Fig. 1). We extract the corresponding triples from ConceptNet based on these two types of correlations. Constructed by these triples, the concept graph is further inputted into a GNN-based encoder, to learn label representations by information propagating and concept correlation representing.

In this article, we propose an end-to-end aerial image classification model, named the Multiple Label Concept Graph (ML-CG), to tackle the aforementioned limitations of current MAIC methods. The proposed ML-CG aims to explore comprehensive semantic correlations between multiple labels. The ML-CG consists of three main components: an image encoder module, a label encoder module, and an image classifier. The image encoder utilizes the pretrained backbone framework to extract representative features of aerial images. The label encoder is a GNN-based model, combining semantic attention and label attention to extract label correlation features from the concept graph extracted from ConceptNet. The image classifier predicts multiple labels by integrating image features and label correlation features.

The main contributions of this article are as follows.

1) Label correlations are learned from both the specific data set labels and the common knowledge graph ecosystem (ConceptNet). To the best of our knowledge, it is the first attempt to explore such label correlations in the MAIC task. The proposed ML-CG builds a concept graph to describe semantic correlations from both the label set for a specific task and the ConceptNet.
2) A novel end-to-end concept attention GNN-based framework, named the Multiple Label Concept Graph (ML-CG), is proposed to extract label correlation features. To the best of our knowledge, no prior work has utilized GNN to exploit label correlations for the MAIC task.
3) Two attention mechanisms, the semantic attention and label attention, are integrated into the GNN framework to better extract the informative correlation features from the input of the concept graph.
4) Experiments are performed on both the commonly used UCM data set and a recently proposed DFC15 data set with high image resolution. The results show the effectiveness of the proposed ML-CG method.

II. Related Work

Recent advances in remote sensing allow easier access to aerial image data. Compared with ground view images, aerial images are with a much larger field of view and generally have more than one object of interest inside. It is natural to move beyond the SAIC task for aerial images. Different from SAIC, the MAIC task aims to map the aerial image features to multiple semantic objects [13].

A. Overview of MAIC Methods

With the recent popularity of deep learning frameworks, various MAIC approaches were developed by designing different network structures. SVM-based models were designed as end-to-end MAIC classifiers for aerial images [14], [15]. Several models based on convolutional neural network (CNN) were designed for specific applications, e.g., unmanned aerial vehicles images [8], and Sentinel-2 imagery acquired over the wet tropics of Australia [16]. Other methods performed image annotation by first partitioning aerial images into instance bags and then classifying the instance bags with different models, e.g., the Mahalanobis distance-based K-medoids approach [17], the CRF model [18], the nonnegative matrix tri-factorization [19], and the bidirectional recurrent neural network architecture [9]. In these methods, image-based graphs were explored to capture global relationships among aerial images [19], [20] and local relationships among spatial regions [21]. In addition, the attention mechanism was first utilized to select the most discriminative class-specific features for label predicting in [22].

Among the abovementioned methods, only a few considered the correlations between objects. Next, we will discuss the related work on exploring label correlations.

B. Label Correlations Beyond the MAIC Task

Label correlations, considered as important supplementary information during the multiple label predicting process, reveal the coexistence of different objects in the same image [23]. In multilabel image classification tasks in computer vision, exploring label correlations is the one top challenging topic and has been investigated in various methods. A probabilistic semantic model was designed to capture relationships between

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TABLE I
SUMMARY OF MAJOR MATHEMATICAL NOTATIONS

Notations	Mathematical meanings
X	the set of n aerial images $\{x_1, x_2, \cdots, x_n\}$
$\mathbf{f}_i$	the m-dimensional feature vector of the image x_i, $\mathbf{f}_i \in R^m$
L	the finite set of c labels (tags or objects) $\{l_1, l_2, \cdots, l_c\}$
$\mathbf{l}_i$	the d-dimensional feature vector of label l_i, $\mathbf{l}_i \in R^d$
$\mathbf{G}$	the label-level feature vectors from label encoder module, $\mathbf{G} \in R^{c\times m}$
$\mathbf{y}_i$	corresponding labels for image x_i, $\mathbf{y}_i = \{y_i^1, y_i^2, \cdots, y_i^c\}$
$\hat{\mathbf{y}}_i$	the predicted result of the image x_i, $\hat{\mathbf{y}}_i \in R^c$
$\mathcal{T}_{CN}$	the ConceptNet triple set
T_{exp}	the explicit triple set
T_{imp}	the implicit triple set
$\mathcal{G}$	the concept graph
H	the neighbor entity set, $H = \{h_1, h_2, \cdots, h_c\}$
h_i	K neighbor nodes of the label node l_i, $h_i = \{h_i^1, h_i^2, \cdots, h_i^K\}$
$\mathbf{h}_i$	feature vectors of neighbor nodes of the label node l_i, $\mathbf{h}_i = \{\mathbf{h}_i^1, \mathbf{h}_i^2, \cdots, \mathbf{h}_i^K\}$
$\tilde{\mathbf{h}}_i$	updated feature vectors of neighbor nodes of the label node l_i, $\tilde{\mathbf{h}}_i = \{\tilde{\mathbf{h}}_i^1, \tilde{\mathbf{h}}_i^2, \cdots, \tilde{\mathbf{h}}_i^K\}$
a_{pq}	the attention score between two neighbor nodes h_i^p and h_i^q
u_{ip}	the attention score between the label node l_i and its one neighbor node h_i^p

low-level features and semantic concepts of images [24]. A low-rank mapping matrix was utilized to signify the label correlation in a more comprehensive and reasonable way [25]. A method was designed based on the graph Laplacian regularization to exploit the label correlation in the local neighborhood [26]. In [10], the GNN was first adapted to model the label correlation in a directly learned structure.

C. Label Correlations for the MAIC Task

While, in the emerging MAIC task, to the best of our knowledge, only limited studies integrated label correlations into the label predicting process. Koda *et al.* [14] exploited interlabel relationships utilizing a structured SVM with the spatial embedding. Zeggada *et al.* [18] utilized a multilabel CRF to encode the cross correlation information between labels within the same image patch. Zhang *et al.* [19] constructed a weighted concept graph to exploit colabeled images. Tan *et al.* [22] built a bidirectional LSTM-based subnetwork to learn label dependencies by sequentially predicting multiple labels. These methods generally defined label correlations by the conditional probability of labels within the data set. Because of this limitation, existing multilabel classifiers for aerial images could be further improved.

D. Motivation

Recently, GNNs provided an alternative way to model the correlations as graphs and can directly deal with graph-structured data in a neural network [27]. Although a GNN provides a more explicit way for evaluating label co-occurrence, it generally updates the node features by the knowledge from one-step neighborhoods around each node. This information propagation relies on the graph structure within the given data set.

To tackle the aforementioned limitations of MAIC approaches, we present a novel end-to-end GNN-based model to explore comprehensive label correlations by integrating the knowledge graph. More specifically, we extract the triples and build a concept graph from both the specific label set and the ConceptNet. ConceptNet is a common knowledge graph with abundant nodes that are explicitly and implicitly related to the labels from the given data set [12]. The concept graph is learned by the proposed GNN module with the incorporated semantic attention and label attention.

III. METHODOLOGY

A. MAIC Task Definition

In the data set, let $X = \{x_1, x_2, \ldots, x_n\}$ represent the aerial image set of n images and $L = \{l_1, l_2, \ldots, l_c\}$ be a finite set of c labels (tags or objects). For an image $x_i \in X$, its corresponding labels are represented as a binary vector $\mathbf{y}_i = \{y_i^1, y_i^2, \ldots, y_i^c\}$, where $y_i^k = 1$ if x_i is annotated with the kth label l_k, $k \in [1, 2, \ldots, c]$; otherwise, $y_i^k = 0$. The goal of MAIC is to train a multilabel image classifier $f : x_i \to \mathbf{y}_i$ that maps the given input image x_i to the output label vector $\mathbf{y}_i$. This can be formulated as a supervised classification problem. For easy reference, Table I lists major notations in the article along with their mathematical meanings.

B. Proposed ML-CG Model for MAIC

The overall ML-CG framework for MAIC is illustrated in Fig. 2, which includes three main components: an image encoder module, a label encoder module, and an image classifier. It is worth mentioning that different backbones can be used for the image encoder component, such as VGGNet, GoogleNet, and ResNet. Here, we only present a particular choice of an image encoder in the proposed ML-CG framework.

Image Encoder Module: We utilize the ResNet network (which is pretrained on ImageNet) to extract the image features. Given each input image x_i, the output feature map with a dimension of $2048 \times 14 \times 14$ from the "*conv*5_*x*" layer is processed by the max-pooling layer to obtain the image-level feature $\mathbf{f}_i \in R^m$ (which is with the dimension of $m = 2048$).

Label Encoder Module: The inputs of the label encoder are the label set L and the ConceptNet triples $\mathcal{T}_{\mathrm{CN}}$. The concept graph (which is described in Section III-C) is built based on

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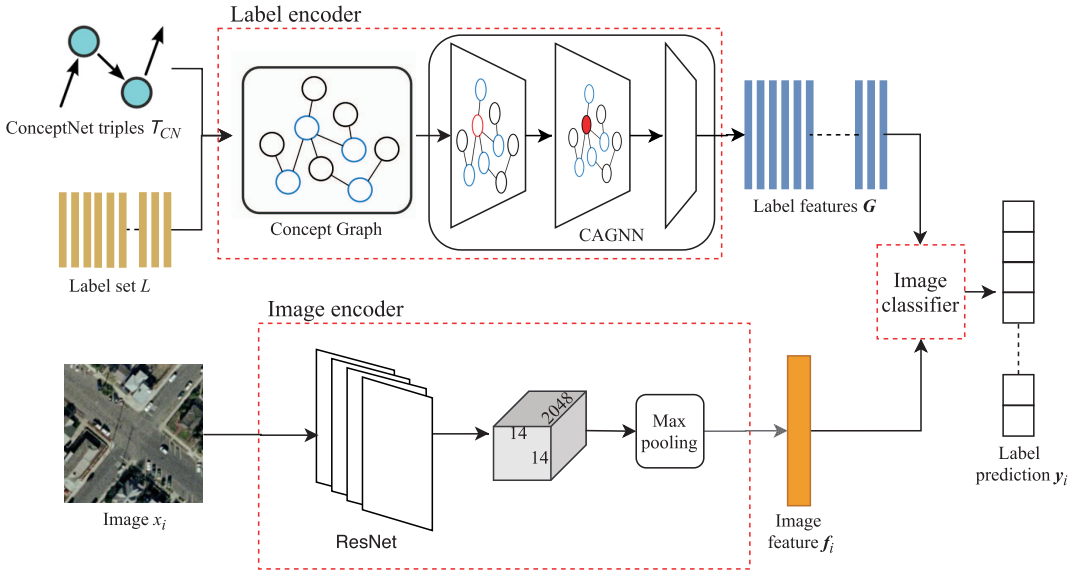


Fig. 2. Proposed ML-CG framework for the MAIC task.

the inputs to get comprehensive label correlations among the objects. Then, the proposed concept attention GNN module (which is described in Section III-D) takes the concept graph as the input and generates the label-level features $\mathbf{G} \in R^{c\times m}$, with $m = 2048$ denoting the dimension of image feature and c being the number of labels.

Image Classifier: For an input image x_i, the final multilabel image classification result $\hat{\mathbf{y}}_i \in R^c$ is predicted by using the image-level feature $\mathbf{f}_i \in R^m$ from the image encoder module and the label-level features $\mathbf{G} \in R^{c\times m}$ from the label encoder module as

$$\hat{\mathbf{y}}_i = sigmoid(\mathbf{G}\mathbf{f}_i^{\mathrm{T}}). \tag{1}$$

Each predicted score $\hat{y}_i^k \in \hat{\mathbf{y}}_i$ represents the confidence of the corresponding label l_k, and each ground-truth value $y_i^k \in \mathbf{y}_i$ represents the presence or absence of the label l_k. For the image x_i, a specific label l_k is predicted as positive if its confidence $\hat{y}_i^k$ is above 0.5. The whole model is trained by the cross-entropy loss function with using the ground-truth vector $\mathbf{y}$ and the predicted $\hat{\mathbf{y}}$ as [10]

$$\mathcal{L} = \frac{1}{c}\sum_{k=1}^{c}[y^k \log(\hat{y}^k) + (1-y^k)\log(1-\hat{y}^k)] \tag{2}$$

where y^k and $\hat{y}^k$ are the kth elements of $\mathbf{y}$ and $\hat{\mathbf{y}}$, respectively, $k \in [1, 2, \ldots, c]$. c is the size of the label set L.

C. Concept Graph Construction

To fully explore label correlations in the MAIC task, we also take advantage of extra semantic knowledge from the ConceptNet triples $\mathcal{T}_{\mathrm{CN}}$. In ConceptNet database, denoting the common entity set as $\mathcal{E}$ and the relation set as $\mathcal{R}$, respectively, we can obtain the whole triples $\mathcal{T}_{\mathrm{CN}} = \{< e_1, r, e_2 > | e_1, e_2 \in \mathcal{E}, r \in \mathcal{R}\}$. A triple in $\mathcal{T}_{\mathrm{CN}}$ represents a correlation between two entities, e.g., < "beach," "is _related _ to," "sea" >. The whole triples $\mathcal{T}_{\mathrm{CN}}$ include both label entities from the label set L (with their correlations) and other common entities in the real-world (with their correlations). We aim to extract sufficient triples from $\mathcal{T}_{\mathrm{CN}}$ to represent the correlations of the label entities and reconstruct a subgraph $\mathcal{G}$ with these extracted triples, which is referred to as a concept graph.

To extract related triples discussed above from $\mathcal{T}_{\mathrm{CN}}$, we define two types of label correlations: explicit correlations and implicit correlations. Triples corresponding to these two label correlations are selected to build the concept graph. The former represents that two label entities adjoin with each other in a triple in $\mathcal{T}_{\mathrm{CN}}$, while the latter represents that two label entities connect indirectly by both adjoining with the third common entity in $\mathcal{T}_{\mathrm{CN}}$. The specific definitions are as follows.

*Definition 1 (*Explicit Correlation*):* For two label entities $l_p \in L$ and $l_q \in L$, if there is a triple $< l_p; r; l_q >$ in $\mathcal{T}_{\mathrm{CN}}$, we call l_p and l_q are explicitly related.

*Definition 2 (*Implicit Correlation*):* For two label entities $l_p \in L$ and $l_q \in L$, if there is a common entity $e \in \mathcal{E} - L$ satisfying that two triples $< l_p; r; e >$ and $< e; r; l_q >$ exist in $\mathcal{T}_{\mathrm{CN}}$, we call l_p and l_q as implicitly related and the common entity e as the third entity.

Based on the above definitions, we can extract two triple sets: the explicit triple set $T_{\mathrm{exp}} = \{< l_p; r; l_q > | l_p, l_q \in L\}$ and the implicit triple set $T_{\mathrm{imp}} = \{< l_p; r; e >, < e; r; l_q > | l_p, l_q \in L, e \in \mathcal{E} - L\}$ from ConceptNet, respectively. For a label entity in the whole extracted triple set $T_{\mathrm{exp}} \cup T_{\mathrm{imp}}$, its neighbor entities can consist of both label entities from the label set L (explicit correlation) and other common entities from ConceptNet $\mathcal{E} - L$ (implicit correlation).

Due to the long tail distribution of entities, some label entities can contain hundreds of neighbor entities, while others only get a few. To reduce the computational complexity and data redundancy, we select the same number of neighbor entities for each label entity. We propose the following: given the whole extracted triples $T_{\mathrm{exp}} \cup T_{\mathrm{imp}}$, the frequency of each entity is counted. For each label entity of interest in our problem,

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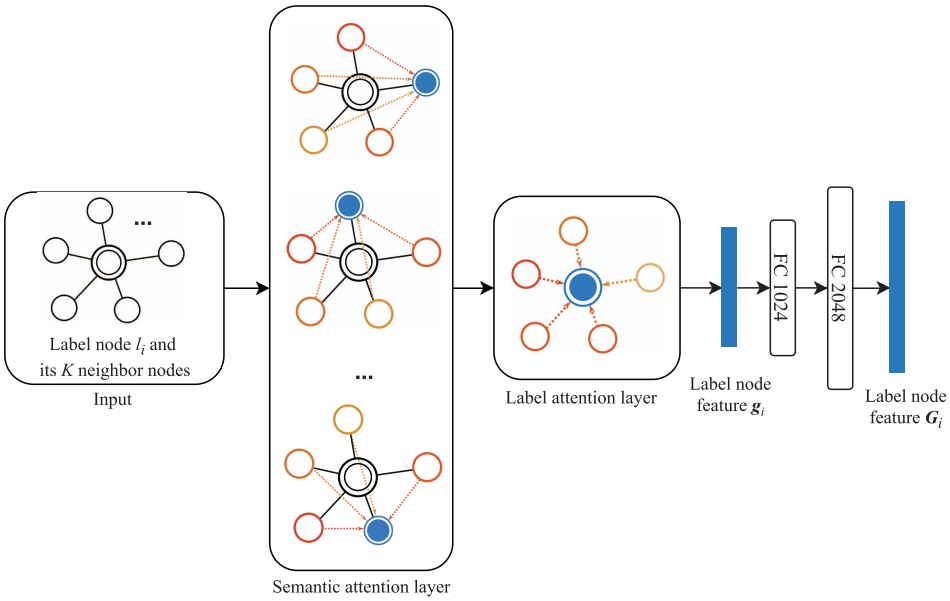


Fig. 3. Proposed CAGNN module with two attention layers. To clarify, we take the subgraph with one label node l_i and its K neighbor nodes as an illustration, which is the input of the CAGNN framework. The aim of the CAGNN is to compute the feature vector $\mathbf{G}_i$ for the label node l_i.

we select up to K high-frequent neighbor entities to keep the information of different label entities compact. Specifically, for label entity $l_i \in L$, its K corresponding neighbor entities are $h_i = \{h_i^1, h_i^2, \ldots, h_i^K\}$. Finally, the concept graph $\mathcal{G}$ is built on label entities (which are from the label set L) and all neighbor entities, denoted as $H = \{h_1, h_2, \ldots, h_c\}$. To clarify, we annotate the entity in triples as nodes in the concept graph.

D. Concept Attention Graph Neural Network

Here, we describe the designed concept attention graph neural network (which is shorted as CAGNN) module in detail. We first represent all nodes in the concept graph $\mathcal{G}$ (built in Section III-C) as d-dimensional feature vectors, where each label node $l_i \in L$ is denoted as $\mathbf{l}_i \in R^d$, and its neighbor node set $h_i \in H$ is denoted as $\mathbf{h}_i = \{\mathbf{h}_i^1, \mathbf{h}_i^2, \ldots, \mathbf{h}_i^K\}$, in which $\mathbf{h}_i^k \in R^d$, $k \in [1, 2, \ldots, K]$. These vectors are then fed into the proposed CAGNN framework.

Although the concept graph can be fed into a traditional GNN directly for the graph feature extraction, there are certain drawbacks of GNNs, as follows. The adjacency matrix is predefined based on conditional probability and cannot be self-adopted during the node feature learning. Also, it is not efficient that all neighbors are treated equally important during information propagation, while we know that the most relevant neighbor information should be concentrated based on the human's understanding. We plan to address such concerns in our proposed approach. More specifically, the attention mechanism can identify the most relevant information by automatically specifying weights to the input sequence. We propose an attention-based CAGNN framework to explore label correlations from concept graphs.

As shown in Fig. 3, we consider one label node l_i and its K neighbor nodes h_i as inputs to the CAGNN module. After two proposed attention layers, *semantic attention layer* and *label attention layer*, the feature of label node l_i is updated as a feature vector $\mathbf{g}_i \in R^d$. Furthermore, the CAGNN outputs the feature $\mathbf{G}_i \in R^n$ of label node l_i.

We now elaborate on the two attention layers.

Semantic Attention Layer: For each label node, the semantic attention layer computes the attention scores for its neighbor nodes around the label node by the *self-attention* mechanism.

Given neighbor node vectors $\mathbf{h}_i = \{\mathbf{h}_i^1, \mathbf{h}_i^2, \ldots, \mathbf{h}_i^K\}$, the initial attention score a_{pq} between two neighbor nodes h_i^p and h_i^q is computed following [28] as

$$a_{\text{pq}} = ReLU(\mathbf{w}^T[W\mathbf{h}_i^p \parallel W\mathbf{h}_i^q]) \tag{3}$$

where $\cdot^T$ denotes the transposition and $\parallel$ represents the concatenation operation. Also, $\mathbf{w} \in R^{2d}$ is a weight vector with ReLU as the activation function. To transform the input neighbor node features into higher-level features, a shared weight matrix $W \in R^{d \times d}$ is applied to every neighbor node.

To compare across different neighbor nodes, the $softmax$ function is applied to normalize the attention score a_{pq} as

$$a_{\text{pq}} = \frac{\exp(a_{\text{pq}})}{\sum_{k=1}^{K} \exp(a_{\text{pk}})}. \tag{4}$$

A higher attention score of one neighbor node indicates more valuable information on which it should be focused on.

The new neighbor node feature $\tilde{\mathbf{h}}_i^p \in R^d$ of the label l_i is obtained by the sum of every attention-weighted neighbor node in $\mathbf{h}_i = \{\mathbf{h}_i^1, \mathbf{h}_i^2, \ldots, \mathbf{h}_i^K\}$ as

$$\tilde{\mathbf{h}}_i^p = \sum_{k=1}^{K} a_{\text{pk}} \mathbf{h}_i^k. \tag{5}$$

To further stabilize the attention mechanism, inspired by the *multihead attention* mechanism in [29], we employ M (which $M = 4$ in our approach) independent attention operations

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TABLE II
NUMBERS OF IMAGES AND CLASSES IN EACH MULTILABEL AERIAL IMAGE DATA SET

Dataset	Image	Class	Training	Testing
UCM	2, 100	17	1, 680	420
DFC15	3, 342	8	2, 674	668

of (5) on the neighbor nodes. Then, the M results are averaged to get the final neighbor node features.

The output of the semantic attention layer for label l_i is its updated neighbor node feature vectors, denoted as $\tilde{\mathbf{h}}_i = \{\tilde{\mathbf{h}}_i^1, \tilde{\mathbf{h}}_i^2, \ldots, \tilde{\mathbf{h}}_i^K\}$. Lately, $\tilde{\mathbf{h}}_i$ together with the label node feature vector $\mathbf{l}_i$ will be treated as the input to the next label attention layer. The label node feature is updated via information propagation from neighbor nodes to the label node.

Label Attention Layer: The label attention layer aims to update the label node feature based on the information from its neighbor nodes. First, the attention score between one label node and its one neighbor node is assigned. Then, the new label node feature is learned by the sum of its attention-weighted neighbor node features.

Specifically, given the input of a label node $l_i \in L$ with its K neighbor node vectors $\tilde{\mathbf{h}}_i = \{\tilde{\mathbf{h}}_i^1, \tilde{\mathbf{h}}_i^2, \ldots, \tilde{\mathbf{h}}_i^K\}$, the attention score between the label node l_i and its one neighbor node h_i^p is calculated referenced to [28] as

$$u_{\text{ip}} = ReLU(\mathbf{w}^T(\mathbf{l}_i \cdot \tilde{\mathbf{h}}_i^p)) \tag{6}$$

where $\mathbf{l}_i \in R^d$ and $\tilde{\mathbf{h}}_i^p \in R^d$ are feature vectors of the label node l_i and neighbor node h_i^p, respectively. $(\cdot)$ represents a dot product operation. $\mathbf{w} \in R^d$ is the weight vector. The same as in (4), the $softmax$ function helps to normalize the attention scores among label nodes.

To integrate information from neighbor nodes, the feature vector of label node l_i is updated as the weighted sum of its K neighbor node features as

$$\tilde{\mathbf{l}}_i = \frac{1}{K}\sum_{k=1}^{K} u_{ik}\tilde{\mathbf{h}}_i^k + \mathbf{l}_i \tag{7}$$

where the updated $\tilde{\mathbf{l}}_i$ is also the output of the label attention layer for label l_i.

To get the final label-level feature $G \in R^{c\times m}$, two fully connected layers (with dimensions of 1024 and 2048, respectively) are performed after the two attention layers.

IV. EXPERIMENTS AND RESULTS

A. Data Sets

To evaluate the performance of the proposed method, we conduct experiments on two aerial image data sets: the UCM multilabel data set [30] and the DFC15 multilabel data set [22]. The statistics of images and classes of each data set are given in Table II. The numbers of images in each class in the total, training, and testing data sets are shown in Fig. 4.

Commonly used in MAIC, the UCM data set consists of 2100 aerial images, and each image has 256 × 256 pixels with a spatial resolution of one foot. There are 17 classes,

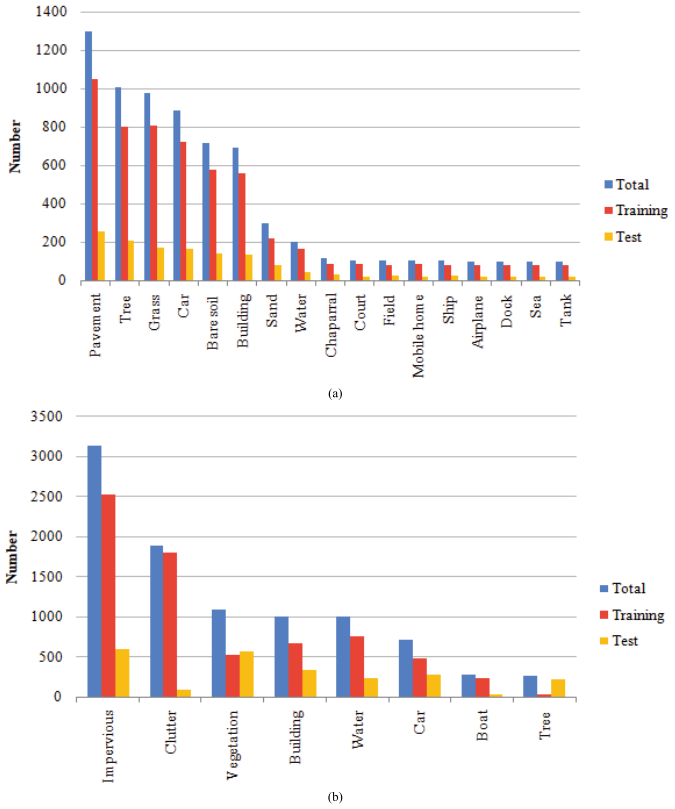


Fig. 4. Numbers of images corresponding to each label in the total data set (blue), training data set (red), and testing data set (yellow). (a) UCM multilabel data set. (b) DFC15 multilabel data set.

including *airplane*, *sand*, *pavement*, *buildings*, *cars*, *chaparral*, *court*, *trees*, *dock*, *tank*, *water*, *grass*, *mobile-home*, *ship*, *bare-soil*, *sea*, and *field*. For network training and testing, we randomly select 80% of sample images evenly from each image category as the training set and use the rest 20% as the testing set.

Compared to the UCM data set, the DFC15 data set is more challenging with an extremely higher spectral resolution of 5 cm and a lower interimage similarity after cropping. There are totally eight labels in the label set, including *impervious*, *water*, *clutter*, *vegetation*, *building*, *tree*, *boat*, and *car*. The number of images is 3; 342. 80% of them are randomly selected as the training set and 20% for the network testing. The DFC15 data set was first published and only used in [22].

B. Evaluation Metrics

Following [10], we calculate both the average example-based precision (EP), recall (ER), F-scores (EF_1 and EF_2), and the average label-based precision (LP), recall (LR), and F-scores (LF_1 and LF_2) for the performance evaluation. For the ablation study, the overall EF_1, LF_1, mAP values, and their values on top-3 selected labels are calculated for fair comparisons.

C. Implementation Details

The network structures of each component in ML-CG are as follows. The image encoder module is composed of the

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TABLE III
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE UCM MULTILABEL DATA SET (%)

Model	EP	ER	EF_1	EF_2	LP	LR	LF_1	LF_2
ResNet-50 [32]	80.86	81.95	81.4	81.73	**88.78**	78.98	83.59	80.76
ResNet-RBFNN [8]	79.92	84.59	82.19	83.61	86.21	83.72	84.95	84.21
CA-ResNet-BiLSTM [22]	77.94	89.02	83.11	86.56	86.12	84.26	85.18	84.63
Image-GCN [21]	75	69	71.86	70.12	76	69	72.33	70.29
MLGCN [10]	79.86	82.10	80.96	81.64	86.42	80.83	83.53	81.89
MLGCN+CG	**82.12**	86.20	84.11	85.35	87.15	84.30	85.70	84.85
ML-CG	81.34	**89.94**	**85.42**	**88.07**	88.53	**89.27**	**88.90**	**89.12**

pretrained Resnet-101 (with the output size of $2048 \times 14 \times 14$ from the "*conv*5_*x*" layer) and a max-pooling layer (with the kernel size of 7×7). The CAGNN in the label encoder module consists of two attention layers (with the output dimension the same as the concept graph input) and two fully connected layers (with the output sizes of 1024 and 2048, respectively). The image classifier is a dot product operation layer (with the output size of the label set).

Concept graph nodes are embedded into n-dimensional feature vectors ($n = 300$) by GloVe (which is pretrained on the Wikipedia data set) [31]. The number of selected neighbor entities is initially set to be $K = 10$. We utilize the SGD as the optimizer of the network and LeakyReLU as the nonlinear activation function. The learning rate is set as 0.03, and the weight decay is 10^{-4}. The network is trained for 20 epochs in total. The momentum is initialized as 0.9, and the minibatch size is initially set to be 8. The network is implemented based on PyTorch on a single NVIDIA GeForce RTX GPU.

D. Experimental Results

In this section, we will present the results of our proposed method on the two data sets respectively. For a fair comparison, we first compare the proposed ML-CG with a standard ResNet framework to demonstrate the effectiveness of exploring label correlations. Then, we compare the proposed method with other methods that utilize the pretrained ResNet as an image feature extractor. In addition, some benchmark comparisons with GNN-based MLC approaches are also carried out. Also, to further verify the effectiveness of the concept graph idea, we then integrated the concept graph component to the MLGCN model in [10], referred to as MLGCN+CG. Furthermore, we conduct ablation studies to demonstrate the necessity of incorporating attention mechanisms and discuss the setting of the parameters.

Results on the UCM Multilabel Data Set: We compare with aforementioned methods, including the ResNet-50 [32], ResNet-RBFNN [8], CA-ResNet-LSTM [22], CA-ResNet-BiLSTM [22], Image-GCN [21], MLGCN [10], and MLGCN+CG. Quantitative results on the UCM multilabel data set are reported in Table III, and the highest scores of different metrics are in bold. It can be seen that the proposed ML-CG achieves superior performance in terms of all studied metrics. For example-based metrics, ML-CG achieves an EP of 81.34% and an ER of 89.94%. It improves EP and ER by 3.4% and 0.92% over CA-ResNet-BiLSTM and 1.48% and 7.84% over MLGCN. In terms of EF_1 and EF_2 metrics that are combined results of both EP and ER, the scores of our model are 85.42% and 88.07%, respectively, which are 2.31% and 1.51% higher than that of the CA-ResNet-BiLSTM model. Although the GCN model from [21] utilized a GNN framework to represent image features, ML-CG outperforms significantly on every metric. Compared with MLGCN [10], the proposed ML-CG increases 4.46% on EF_1 and 6.43% on EF_2, providing a stronger ability on both precision and recall.

For label-based metrics, 88.53% and 89.27% are obtained for LP and LR by the proposed method, representing a corresponding improvement of 2.41% and 5.03% over CA-ResNet-BiLSTM. For the proposed method, the scores of LF_1 and LF_2 reach 88.9% and 89.12%, which are 3.72% and 4.49% higher than that of CA-ResNet-BiLSTM. Although 0.25% lower than that of ResNet-50 on LP, our model is 10.29% higher on LR, showing that our model can achieve high recall while maintaining precision. Also, in comparison with MLGCN [10], ML-CG achieves 5.37% higher on LF_1 and 7.23% higher on LF_2.

The improvements by MLGCN+CG show the effectiveness of the concept graph. During label predicting, the concept graph provides significant complementary information from label correlations. In summary, our comparison results demonstrate that the proposed ML-CG can achieve better performance by exploring semantic correlations between labels in the knowledge graph.

Results on the DFC15 Multilabel Data Set: We compare with aforementioned methods, including the ResNet-50 [32], ResNet-RBFNN [8], CA-ResNet-LSTM [22], CA-ResNet-BiLSTM [22], MLGCN [10], and MLGCN+CG. Experimental results on the DFC15 multilabel data set are shown in Table IV. It can be seen that our ML-CG model outperforms significantly on every metric, demonstrating the effectiveness of the concept graph extension and the proposed CAGNN component. In terms of example-based metrics, the proposed ML-GC method obtains 95.68% and 96.09% on precision and recall respectively, much higher than that of CA-ResNet-BiLSTM and MLGCN. EF_1 and EF_2 of ML-CG are 95.89% and 96.01%, respectively, which improves the state-of-the-art results by 2.25% and 2.30%, respectively. In terms of label-based metrics, the proposed ML-CG model improves LR by 3.13% and reaches 94.28%. Meanwhile, LF_1 is 93.96%, and LF_2 is 93.76%, which shows a more balanced performance of

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TABLE IV
PERFORMANCE COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE DFC15 MULTILABEL DATA SET (%)

Model	EP	ER	EF_1	EF_2	LP	LR	LF_1	LF_2
ResNet-50 [32]	84.89	75.64	80.00	77.33	81.5	59.99	69.11	63.33
ResNet-RBFNN [8]	82.64	78.76	80.65	79.51	72.01	69.85	70.91	70.27
CA-ResNet-LSTM [22]	85.66	75.84	80.45	77.62	83.83	60.05	69.97	63.66
CA-ResNet-BiLSTM [22]	91.93	79.12	85.05	81.39	94.35	62.35	75.08	66.89
MLGCN [10]	93.52	93.76	93.64	93.71	91.15	90.02	90.58	90.24
MLGCN+CG	94.46	94.56	94.51	94.54	92.86	92.08	92.47	92.23
ML-CG	**95.68**	**96.09**	**95.89**	**96.01**	**94.28**	**93.63**	**93.96**	**93.76**

precision and recall. Our performances improve around 20% and 30% compared to CA-ResNet-BiLSTM. Also, compared with the MLGCN, it can be seen that the proposed ML-CG improves a lot on both example- and label-based metrics. For EF_1 and LF_1 on the DFC15 data set, ML-CG is 1.38% and 1.49% higher than that of the MLGCN model. Besides, MLGCN+CG improves the MLGCN by increasing 0.87% on EF_1 and 1.89% on LF_1, indicting the benefit from the concept graph for label predicting. To sum up, the proposed ML-GC model achieves significant improvements in the newly proposed data set with high image resolutions. Since this data set is more practically challenging, our results validate that the proposed model is promising in real-world MAIC applications.

E. Concept Graph for Label Nodes

In this section, we visualize the results on the concept graph that explores comprehensive label correlations. The proposed ML-CG framework models label correlations in the concept graph. The ML-CG first extracts explicitly and implicitly correlated nodes from the ConceptNet. Then, the proposed model crops the long tail distribution by selecting the top K most-frequent neighbor entities for each label node of interest.

Fig. 5 shows two concept subgraph for the label node *water* and the label node *cars*. In Fig. 5, the red nodes represent the labels from our specific label set L (which is given for each specific data set), and gray nodes are other common entities extracted from the ConceptNet. We can see from the concept graph in Fig. 4E that the label node *cars* is explicitly correlated with ten neighbor nodes, including *vehicle*, *ferry*, *bridge*, *glass*, *findoutside*, *stick*, *road*, *windows*, *city*, and *wheels*. The label node *cars* is implicitly correlated with nine label nodes, including *ships*, *airplane*, *mobile home*, *court*, *buildings*, *pavement*, *trees*, *grass*, and *chaparral*, while the label node *water* in Fig. 4E has direct connections with two label nodes from our label set L, including *trees* and *sea*, as well as eight other common nodes from the ConceptNet, including *rock*, *surface of earth*, *ground*, *ocean*, *vessel*, *sea plane*, *land*, and *earth*.

On the other hand, the number of two-step paths from one label node to other red label nodes reflects the correlations between them. In other words, the more paths exist between the two nodes, the more correlated the two nodes are. For example, in Fig. 4E, there are three paths from label node *cars* to label node *ship*, while there is only one path from label node *cars* to label node *trees*, indicating that *cars* is more correlated with *ship* than *trees*. Also, in Fig. 4E, the number of implicit paths from *water* and *sea* is five, indicating that the two label nodes are closely correlated with each other. In our proposed framework, the concept graph is then fed into the CAGNN module and is learned by the two attention layers.

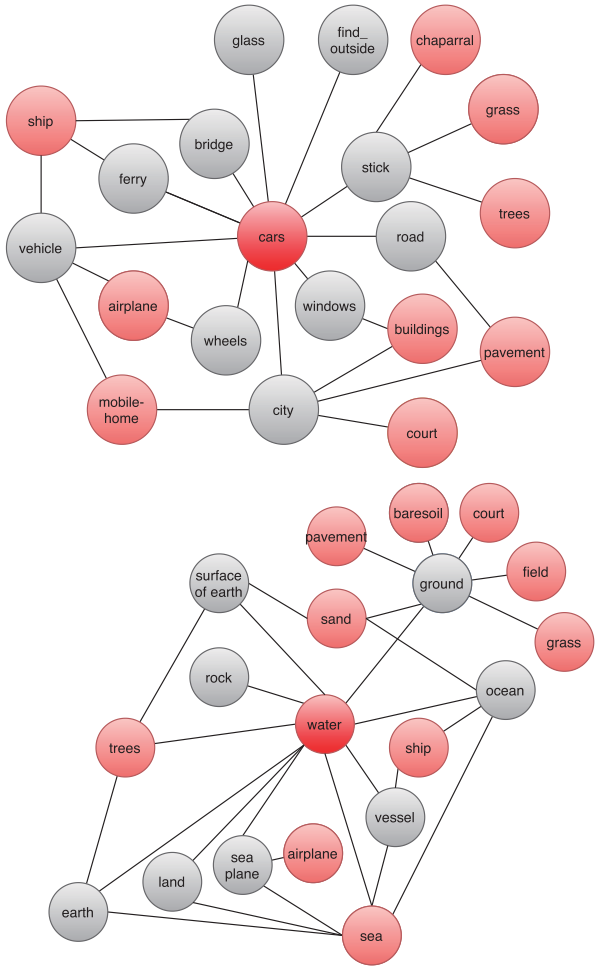


Fig. 5. Concept subgraphs for the label node *car* and the label node *water* from the UCM multilabel data set. (a) Label node *car* (bright red) and its ten most frequent neighbor nodes and some implicitly correlated label nodes (red). (b) Label node *water* (bright red) and its ten most frequent neighbor nodes and some implicitly correlated label nodes (red). The nodes in ConceptNet are indicated in gray, and the nodes in our specific label set L are in red.

F. Ablation Study

Attention Mechanism: To verify the necessity of two attention mechanisms in the CAGNN module, we perform ablation

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TABLE V

PERFORMANCE COMPARISONS WITH DIFFERENT ATTENTION MECHANISMS IN OUR MODEL ON THE UCM MULTILABEL DATA SET (%)

Model	All			Top 3	
	EF_1	LF_1	mAP	EF_1	LF_1
CG-GNN	82.98	87.27	92.83	76.38	**83.50**
CG-A_1GNN	84.54	87.82	94.10	77.26	82.36
CG-A_1A_2GNN	**85.42**	**88.90**	**94.82**	**77.77**	83.11

TABLE VI

PERFORMANCE COMPARISONS WITH DIFFERENT ATTENTION MECHANISMS IN OUR MODEL ON THE DFC15 MULTILABEL DATA SET (%)

Model	All			Top 3	
	EF_1	LF_1	mAP	EF_1	LF_1
CG-GNN	94.67	92.85	97.56	91.25	86.73
CG-A_1GNN	95.09	93.16	98.46	91.47	87.87
CG-A_1A_2GNN	**95.89**	**93.96**	**98.61**	**91.99**	**87.94**

studies of attention layers. We compare the EF_1, LF_1, and mAP metrics on both the overall results and top-3 precision results. The results are shown in Tables V and VI. *CG-GNN* represents that the ML-CG model updates the nodes by direct summation of the neighbors with no attention consideration. *CG-A_1GNN* represents that only the label attention layer is used in ML-CG. *CG-A_1A_2GNN* represents that the CAGNN with both semantic attention and label attention in ML-CG. It can be noted that, by integrating the attention mechanism layer by layer, the performance improves gradually. For the UCM data set, after adding the label attention layer, *CG-A_1GNN* increases the mAP by 1.27% compared with *CG-GNN* without attention, indicating the necessity of the label attention layer. *CG-A_1A_2GNN* achieves the best performance on almost all metrics, demonstrating the effectiveness of two attention layers in our model.

Another observation of interest is that, when only utilizing the *CG-GNN* module, the proposed model achieves EF_1 of 82.98% and LF_1 of 87.27% on UCM and EF_1 of 94.67% and LF_1 of 92.85% on DFC15. The *CG-GNN* module without attention consideration still outperforms the existing methods. The improvement could be due to the consideration of extensive label correlations from the ConceptNet based on two explicit and implicit correlations, as presented in Section III-C and illustrated in Section IV-E.

Number of Neighbor Nodes K: To reduce the computation complexity and redundancy, we select the top-K most frequently occurred neighbor nodes to construct the concept graph. To explore the proper number of neighbor nodes for one label node, the value of K is selected in a set of {2, 4, 8, 10}. The mAP results on different K values are shown in Fig. 4F. For the UCM data set, the mAP results are significantly different when K changes and get the best with $K = 10$ neighbor nodes. In contrast, the results do not change much on the DFC15 data set with the different number of neighbor nodes. The reason may be that the number of labels on the

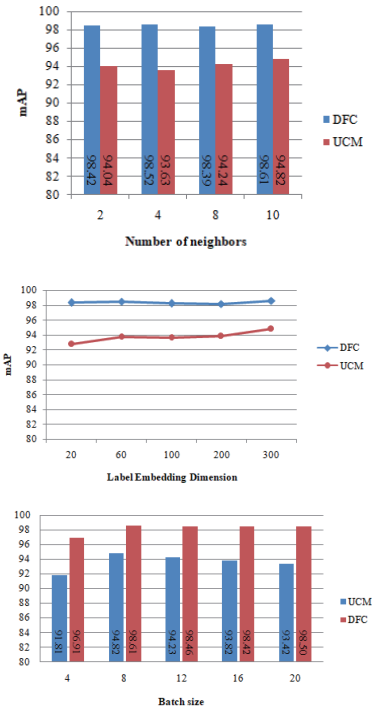


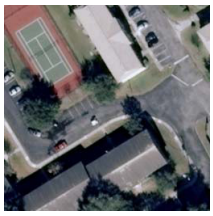
Fig. 6. Results with different parameters. (a) mAP as a function of the number of neighbor nodes K on UCM (red) and DFC15 (blue) data sets. (b) mAP results on different dimensions n of node feature vectors on UCM (red) and DFC15 (blue) data sets. (c) mAP results with different batch sizes on UCM (blue) and DFC15 (red) data sets.

DFC15 label set is only 8, much less than that of the UCM label set, which is 17. The diversity scope of the topology among eight label nodes on DFC15 is limited, so the results do not change dramatically when adding neighbor nodes to the concept graph. The proposed ML-CG model performs well even with the small number of neighbor nodes on the concept graph, indicating the strong inference ability of the CAGNN with semantic and label attention layers.

Node Feature Vectors' Dimensions n: We represent the node features of the concept graph by utilizing the output of the pretrained Glove embedding model with an initial dimension of $n = 300$. To explore the effects of different node embedding dimensions on the model performance, in this part, we evaluate the mAP performance of the proposed ML-CG model under several node embedding dimensions. We set the dimension of node features as {100, 150, 200, 250, 300}, and the results are shown in Fig. 4F. The node feature vector dimension can impact the final precision of the model. For the DFC15 multilabel data set, the mAP varies slightly on different dimensions and achieves the best on the dimension of 300. For the UCM multilabel data set, the mAP increases obviously when the dimension varies from 100 to 300 and the best performance is achieved at the dimension of 300. The intuitive explanation for these results can be that, if the node feature dimension is too small, the features of nodes in the concept graph will not be sufficiently represented by the model; if the node embedding dimension is too large, the computation cost can be high.

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TABLE VII
EXAMPLE PREDICTIONS ON THE UCM MULTILABEL DATA SET

Images in UCM multi-label dataset	 (a)	 (b)	 (c)	 (d)
Ground Truth	bare-soil, buildings, cars, court, grass, pavement, trees	bare-soil, buildings, cars, grass, pavement, trees	grass, trees, water	buildings, pavement
Predictions	bare-soil, buildings, cars, court, grass, pavement, trees	bare-soil, buildings, cars, grass, pavement, trees	grass, pavement, sand, trees, water	bare-soil, buildings, grass, pavement, tanks

Effects of Different Batch Sizes: We show the mAP performance with different values of the batch size on two data sets. As our model is trained by a batch of samples, the size of the batch in every epoch can affect the performance and the convergence of the model. We select the value of batch size from a set of {4, 8, 12, 16, 20} and draw the histogram to show the results, as depicted in Fig. 6. The figure shows that the mAP performance varies on UCM and DFC15 multilabel data sets when using different batch sizes. We can see that, when the batch size is set to be 8, the proposed ML-CG model yields the best results on both data sets.

V. DISCUSSION

In this section, to further validate the effectiveness of our model, we conduct two kinds of qualitative analyses on the proposed ML-CG model, including the case study and the label-specific feature visualization. First, the case study shows examples of several image prediction results from our proposed ML-CG model on the UCM multilabel data set, as illustrated in Table VII. Then, the label-specific feature visualization reveals the most task-specific patches of images related to positive labels, as shown in Table VIII.

A. Case Study

We note that the proposed ML-CG generally works well and can classify images accurately even for those with a large number of labels. For instance in Table VII, image (a) has up to seven ground-truth labels, including *bare-soil*, *buildings*, *cars*, *court*, *grass*, *pavement*, and *trees*. All these labels are identified correctly in our multilabel image classification results by ML-CG. Image (b) has up to six labels, including *bare-soil*, *buildings*, *cars*, *grass*, *pavement*, and *trees*; these labels are also predicted accurately by the ML-CG model.

We also note that, since ML-CG holds a strong inference ability for multilabel prediction, it can even identify objects that are not included in the ground-truth labels. For example, in Table VII, image (c) has only three ground-truth labels, including *grass*, *trees*, and *water*, while, in our prediction results, the proposed ML-CG also identifies *pavement* and *sand* as positive labels, which are missed in the given ground-truth labels. Also, for image (d), other than correctly predicting two ground-truth labels, the ML-CG provides more comprehensive understandings of this image, e.g., correctly identifying additional *bare-soil*, *grass*, and *tank* labels. The intuitive reason could be the following: different from previous methods where label correlations are often explored only within the acquired data set (i.e., the label set L), the proposed ML-CG can extract more implicit label correlations by using the common knowledge ConceptNet and further explore attention-weighted features of the labels. Therefore, more possible options can be provided when choosing related labels (from both L and ConceptNet) during label predicting.

B. Label-Specific Feature Visualization

To further study the effectiveness of the proposed ML-GC model, we conduct the label-specific feature visualization analysis on UCM images. The qualitative experiments reveal the visual interpretation of our model, and the results are illustrated in Table VIII. We can see from Table VIII that the regions with the most specific semantic information related to positive labels are focused and highlighted in the feature maps. Meanwhile, the patches with less label-correlated information show less activation and attention.

C. Comparison of Model Efficiency

To further evaluate the performance of the proposed ML-CG, we study its model efficiency. Specifically, the calculation complexity and parameters of our model are investigated. We have utilized the $flops_counter$ package in PyTorch and calculated the floating-point operations (FLOPs) of our model directly [33]. As reported in Table IX, the standard backbone (the original ResNet-101 [32]) and the aforementioned method (MLGCN [10]) are investigated for comparison. Also, we compare with the backbone with only one fully connected layer ($Backbone + FC$ in Table IX). In terms of MACs, our

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TABLE VIII

EXAMPLE LABEL-SPECIFIC FEATURES OF A FEW SAMPLES SELECTED FROM THE UCM MULTILABEL DATA SET

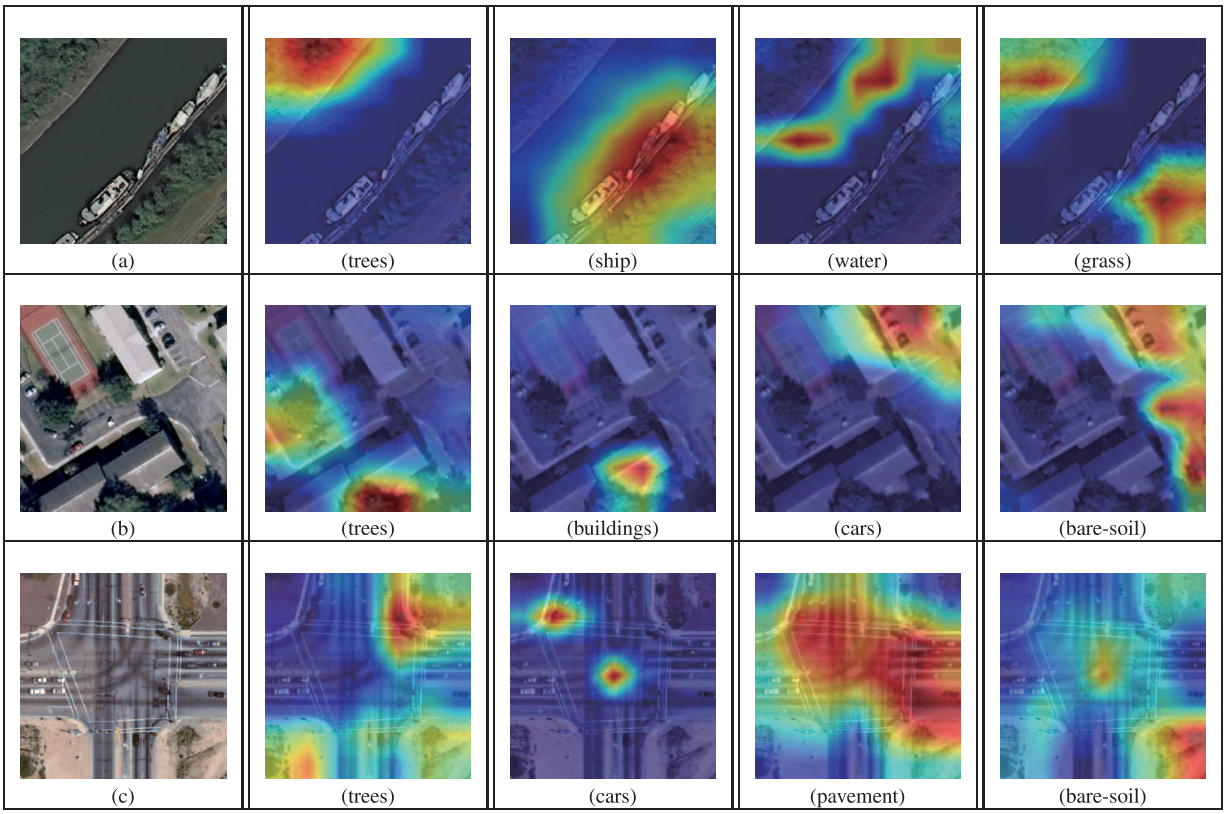


TABLE IX

COMPARISONS OF MODEL EFFICIENCY

Model	FLOPs ($\times 10^9$)	Parameters ($\times 10^6$)
ResNet-101 [32]	7.85	44.55
MLGCN [10]	7.83	42.50
Backbone+FC	7.84	43.12
ML-CG	7.87	44.90

model is only 0.03 higher than that of $Backbone + FC$ and 0.04 higher than that of MLGCN [10]. In terms of parameters, our model has 44.90M of parameters, slightly higher than that of $Backbone + FC$ and MLGCN [10]. The comparison of model efficiency shows that the proposed ML-CG does not require much more resources compared with other models.

VI. CONCLUSION

In this article, we propose the multiple label concept graph (ML-CG) approach based on the concept attention graph neural network for multilabel aerial image classification. The advantages of the proposed ML-CG are as follows. First, ML-CG can better make use of the underlying semantic correlations between image labels by exploiting the real-world knowledge ecosystem, which is a comprehensive supplement for multiple label prediction. Second, ML-CG has better flexibility and robustness. Based on the designed concept attention GNN, ML-CG can be adapted into practical multilabel aerial image classification applications. The effectiveness of ML-CG is demonstrated on both the commonly used UCM data set and the more challenging DFC15 data set. On both data sets, ML-CG outperforms the state-of-the-art methods in terms of both example- and label-based metrics. In the future, we will explore label correlations by utilizing the complementary cues from images to better tackle the MAIC task.

REFERENCES

[1] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 140, pp. 20–32, Jun. 2018.

[2] D. Wen, X. Huang, H. Liu, W. Liao, and L. Zhang, "Semantic classification of urban trees using very high resolution satellite imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1413–1424, Apr. 2017.

[3] L. Mou and X. Xiang Zhu, "IM2HEIGHT: Height estimation from single monocular imagery via fully residual convolutional-deconvolutional network," 2018, *arXiv:1802.10249*. [Online]. Available: http://arxiv.org/abs/1802.10249

[4] Q. Weng, Z. Mao, J. Lin, and X. Liao, "Land-use scene classification based on a CNN using a constrained extreme learning machine," *Int. J. Remote Sens.*, vol. 39, no. 19, pp. 6281–6299, Oct. 2018.

[5] A. Romero, C. Gatta, and G. Camps-Valls, "Unsupervised deep feature extraction for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1349–1362, Mar. 2016.

[6] G. Tsagkatakis and P. Tsakalides, "Deep feature learning for hyperspectral image classification and land cover estimation," in *Proc. ESA Symb.*, 2016, pp. 1–6.

[7] T. Zhang, W. Yan, J. Li, and J. Chen, "Multiclass labeling of very high-resolution remote sensing imagery by enforcing nonlocal shared constraints in multilevel conditional random fields model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 7, pp. 2854–2867, Jul. 2016.

[8] A. Zeggada, F. Melgani, and Y. Bazi, "A deep learning approach to UAV image multilabeling," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 694–698, May 2017.

[9] G. Sumbul and B. Demir, "A novel multi-attention driven system for multi-label remote sensing image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 5726–5729.

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[10] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5177–5186.
[11] Z. Wang, J. Zhang, J. Feng, and Z. Chen, "Knowledge graph embedding by translating on hyperplanes," in *Proc. AAAI*, 2014, pp. 1112–1119.
[12] R. Speer, J. Chin, and C. Havasi, "Conceptnet 5.5: An open multilingual graph of general knowledge," in *Proc. AAAI*, 2017, pp. 4444–4451.
[13] K. Chen, P. Jian, Z. Zhou, J. Guo, and D. Zhang, "Semantic annotation of high-resolution remote sensing images via Gaussian process multi-instance multilabel learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1285–1289, Nov. 2013.
[14] S. Koda, A. Zeggada, F. Melgani, and R. Nishii, "Spatial and structured SVM for multilabel image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5948–5960, Oct. 2018.
[15] A. Dixit, N. Hedge, and B. E. Reddy, "Texture feature based satellite image classification scheme using SVM," *Int. J. Appl. Eng. Res.*, vol. 12, no. 13, pp. 3996–4003, 2017.
[16] I. Shendryk, Y. Rist, R. Lucas, P. Thorburn, and C. Ticehurst, "Deep Learning–a new approach for multi-label scene classification in planetscope and Sentinel-2 imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1116–1119.
[17] X. Wang, X. Xiong, and C. Ning, "Multi-label remote sensing scene classification using multi-bag integration," *IEEE Access*, vol. 7, pp. 120399–120410, 2019.
[18] A. Zeggada, S. Benbraika, F. Melgani, and Z. Mokhtari, "Multilabel conditional random field classification for UAV images," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 3, pp. 399–403, Mar. 2018.
[19] J. Zhang, J. Zhang, T. Dai, and Z. He, "Exploring weighted dual graph regularized non-negative matrix tri-factorization based collaborative filtering framework for multi-label annotation of remote sensing images," *Remote Sens.*, vol. 11, no. 8, p. 922, Apr. 2019.
[20] Q. Tan, Y. Liu, X. Chen, and G. Yu, "Multi-label classification based on low rank representation for image annotation," *Remote Sens.*, vol. 9, no. 2, p. 109, Jan. 2017.
[21] N. Khan, U. Chaudhuri, B. Banerjee, and S. Chaudhuri, "Graph convolutional network for multi-label VHR remote sensing scene recognition," *Neurocomputing*, vol. 357, pp. 36–46, Sep. 2019.
[22] Y. Hua, L. Mou, and X. X. Zhu, "Recurrently exploring class-wise attention in a hybrid convolutional and bidirectional LSTM network for multi-label aerial image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 188–199, Mar. 2019.
[23] M.-L. Zhang and Z.-H. Zhou, "A review on multi-label learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 8, pp. 1819–1837, Aug. 2014.
[24] I. Ben Rejeb, S. Ouni, W. Barhoumi, and E. Zagrouba, "Fuzzy VA-files for multi-label image annotation based on visual content of regions," *Signal, Image Video Process.*, vol. 12, no. 5, pp. 877–884, Jul. 2018.
[25] J. Wu, A. Guo, V. S. Sheng, P. Zhao, and Z. Cui, "An active learning approach for multi-label image classification with sample noise," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 32, no. 3, Mar. 2018, Art. no. 1850005.
[26] X. Yang, Y. Zhou, Q. Zhu, and Z. Wu, "Joint graph regularized extreme learning machine for multi-label image classification," *J. Comput. Methods Sci. Eng.*, vol. 18, no. 1, pp. 213–219, Feb. 2018.
[27] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. Van Den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in *Proc. Eur. Semantic Web Conf.* Cham, Switzerland: Springer, 2018, pp. 593–607.
[28] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
[29] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liü, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*. [Online]. Available: http://arxiv.org/abs/1710.10903
[30] B. Chaudhuri, B. Demir, S. Chaudhuri, and L. Bruzzone, "Multilabel remote sensing image retrieval using a semisupervised graph-theoretic method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1144–1158, Feb. 2018.
[31] J. Pennington, R. Socher, and C. Manning, "Glove: Global vectors for word representation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1532–1543.
[32] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[33] P. Molchanov, S. Tyree, T. Karras, T. Aila, and J. Kautz, "Pruning convolutional neural networks for resource efficient inference," 2016, *arXiv:1611.06440*. [Online]. Available: http://arxiv.org/abs/1611.06440



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